Product Support Bulletin

Upgrade Neptune 2 feeder door latch

PSB #: PSB-A11-03
System: H9000
Date: February 2011

Overview

ECN008842 was released to upgrade the existing feeder door latch to a new mechanism in order to improve the performance and reliability of the feeder door.

Systems Affected

All Neptune 2 systems shipped before May 2011 are potentially affected.

Compatibility

There are no compatibility issues.

Impact

The feeder door will be open for approximately 10 minutes while executing this Product Support Bulletin.

Technical Description

This procedure improves the feeder door elevator interlock's reliability and safety by: <QUESTION 1>

- Replacing the spike with a rubber pusher
- Adding a slider assembly to the interlock

Tool List <QUESTION 2>

Part Number	Description	Quantity
470-471-09	SCRW CP SKT HD M6X1.0P/30MM SS	2
601-881-01	SPRING COMP 7/32ODx1"L MUSIC WR	1
612-434-00	ME FEEDER DOOR ELEVATOR INTERLOCK SLIDER HOUSING	1
612-436-00	ME FEEDER DOOR ELEVATOR INTERLOCK SLIDER	1
606-631-05	PUSHER M5x12DI Ax16MML, SS-POLYURETHN	1
	M5 hex nut driver	1
	10mm wrench	1
	8mm wrench	1
	M4 Wrench	1

	M3 Wrench	1

Installation Instructions <QUESTION 3>

To replace the spike assembly with the rubber pusher:

1. Lift the raise tote plate free of the feeder door. <QUESTION 4>
2. Use an M4 Allen Wrench to loosen the set screw securing the feeder door.
3. Use an M4 Allen Wrench to remove the screw holding the feeder door closed.
4. Use an M3 Allen Wrench to remove the final screw holding the feeder door closed.
5. Open the feeder door. <QUESTION 5>
6. Use an M5 hex nut driver to remove the two M5 nuts securing the spike assembly to the feeder door.
7. Remove the two lock washers from the spike assembly.
8. Remove the two washers from the spike assembly.
9. Gently pull the spike assembly away from the feeder door to remove it. <QUESTION 6>
10. Use an M5 hex nut driver to screw an M5 nut onto the bottom stud that formerly held the spike assembly.
11. Screw the rubber pusher onto the bottom stud. <QUESTION 7>

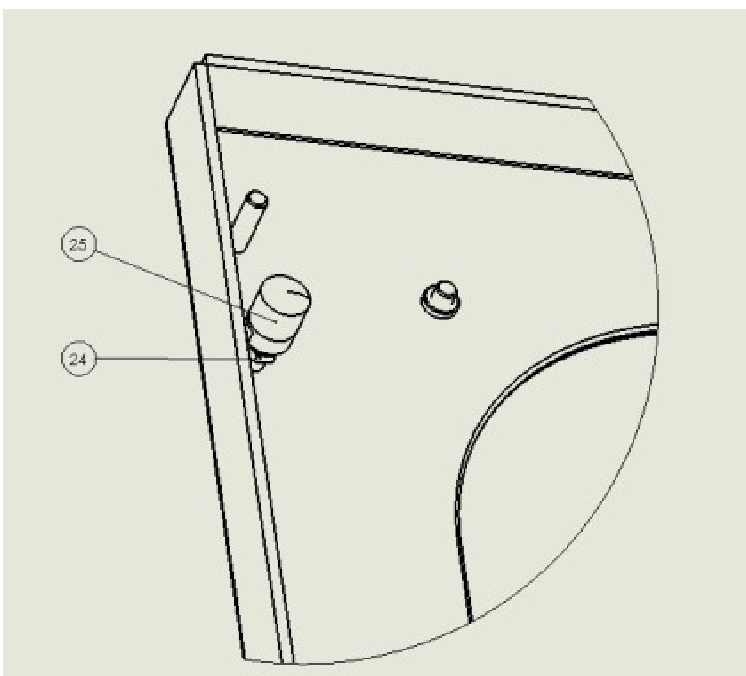

Figure 1: Properly installed rubber pusher.

To install the slider assembly into the interlock:

1. Use an M5 wrench to remove the two bottom screws securing the outer plate of the latch block housing.

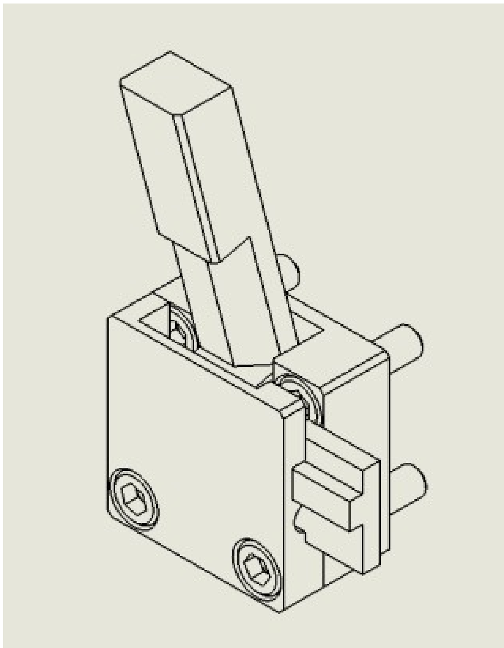

Figure 2: Assembled latch block.

2. Insert the spring (601-881-01) into the hole in the slider.
3. Insert the protruding end of the spring into the hole in the housing, so that the slider gently rests on top of the inner ledge of the housing.
4. Insert two M6 socket head cap screws into the two holes at the bottom of the latch block housing.

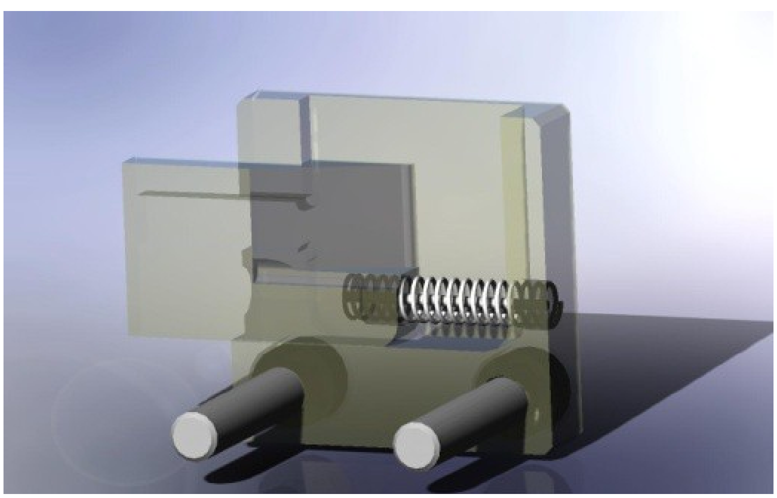

Figure 3: Slider assembly installed into the latch block housing, with M6 socket head cap screws inserted.

5. Using one hand to prevent the slider assembly from coming loose from the latch block housing, align the latch block housing with the feeder door such that the two M6 socket head cap screws

are aligned with their holes in the feeder door. <QUESTION 8>

6. Use an M5 wrench to tighten both screws to a maximum torque of 29 in-lbs. <QUESTION 9>

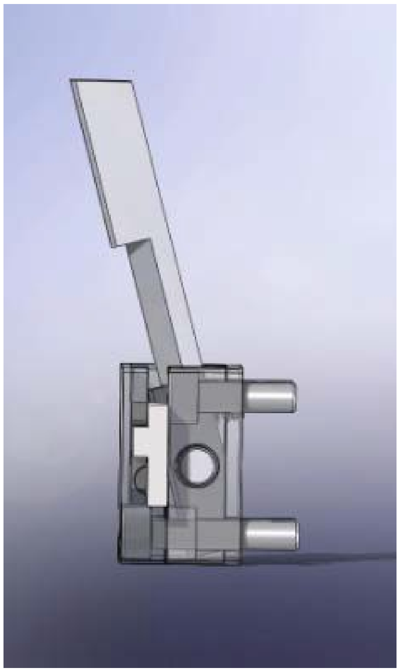

Figure 4: Reassembled latch block assembly.

7. Use a 10mm wrench to grip the flat sides of the rubber pusher, and an 8mm wrench to grip the M5 nut. Jam the pusher in place by twisting the rubber pusher clockwise and the nut counter-clockwise to a maximum torque of 29 in-lbs.

To test the slider assembly and ensure it has been installed properly:

1. Gently press the slider towards the latch block housing. If installed properly, the slider will return to its original position when released.

Figure 5: Testing the slider.

2. Slowly pull and release the interlock latch. If installed properly, the slider will retract when the

latch is pulled, and return to its original position when released. <QUESTION 10>

3. With the feeder door open, make sure the interlock latch rests on an angle to prevent the elevator from lowering while the door is open, such that the upper inner corner of the interlock latch rests at least 20 mm from the feeder wall.

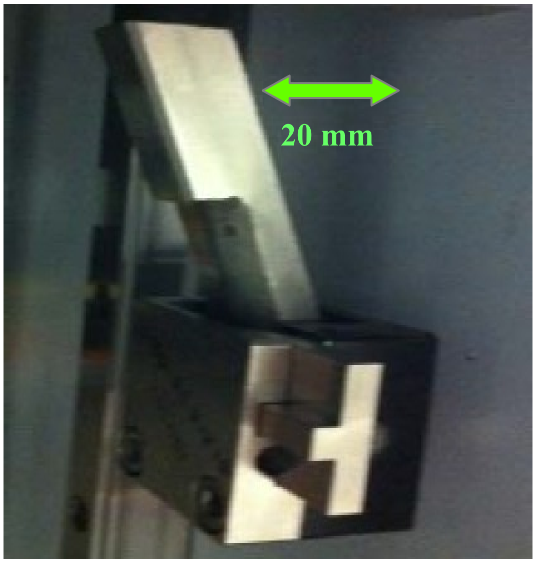

Figure 6: Latch positioning while the feeder door is open.

4. Close the feeder door. While the feeder door is closed, make sure the latch is parallel to the feeder wall. If the latch is not parallel to the feeder wall while the feeder door is closed, adjust the pusher so that it protrudes enough to push the slider in. <QUESTION 11> <QUESTION 12> <QUESTION 13>

5. With the feeder door closed, make sure the slider protrudes less than 2mm from the latch block housing.

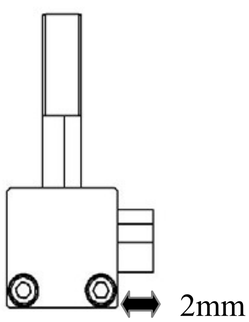

Figure 7: Slider positioning while the feeder door is closed.

EDITOR QUESTIONS

Below are a list of the questions I have for the engineering team regarding this procedure, noted as <QUESTION #> throughout this document.

1. How does the replacement slider assembly improve the performance of the feeder door?
2. Is the tool list a list of all the items required to perform this procedure, or a list of all items included with some sort of kit provided to perform the procedure? What qualifies/disqualifies an item from being included in this table?
   Also, are the techs actually using this procedure familiar with part numbers and the engineering shorthand description of each item, or do they prefer each item presented in English? I also changed "rubber bumper" in the procedure to "rubber pusher" to match the description in the tool list, since it should be referred to with a consistent term.
3. Is the Neptune 2 a powered device, and does this procedure require the unit to be de-powered to be performed safely?
4. How is the Raise Tote attached or secured in front of the feeder door?
5. How does the feeder door open? Is there a latch or lever that needs to be utilized, or can the door be simply pulled open?
6. After removing the 2 nuts and 4 washers securing the spike assembly to the feeder door, can the spike assembly be simply pulled off of the door?
7. Can the rubber pusher be hand-screwed onto the bottom stud, or is a tool required? (before using a 10mm wrench to jam the pusher into place, which happens after installing the interlock assembly.
8. Can you offer any elaboration on the step of "install the interlock upgrade onto existing latch block." (bullet 3 of step 5 in the original doc)? I thought installing the interlock upgrade onto the latch block was what was described in the first 2 bullet points of step 5. Unless it's trying to tell the reader to place the upgraded assembly against the part of the latch block housing still attached to the feeder door so that the screw holes are aligned, before tightening the screws in bullet 4.
9. The part description at the top says that there are 2 M6 screws included, and I'm assuming these are the screws described in step 5 > bullet 4 since the part numbers match. But that bullet point says to use an M5 wrench on the M6 screws- is this correct?
10. What action engages the latch? Pull or push?
11. How can you check the latch while the feeder door is closed? Is the door transparent? Is there a tool you can insert that will check the position of the latch while the door is closed? Is the latch visible through the feeder window (or is the feeder window accessible?)
12. How does one adjust the pusher? Tighten/loosen it with a 10mm wrench? Is this possible after it has been jammed (per step 7 of installing the slider assembly), or will a new pusher need to be installed?
13. Should adjusting the pusher post-installation be its own documented procedure here? How common of an issue is this?

14. Is it Teradyne style to include the part number in parenthesis every single time a numbered part is mentioned?